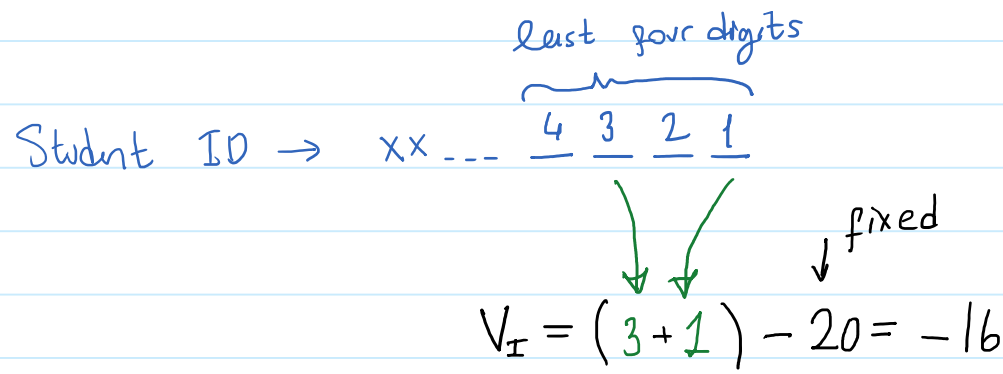
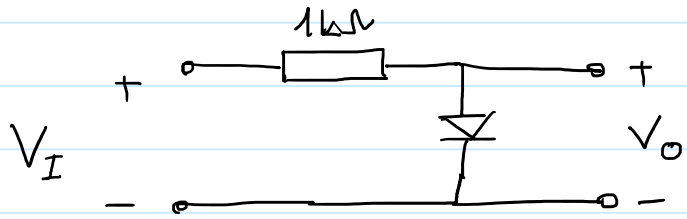
Q1



$V_I =$? Write only the answer, no calculations!!!

Q2



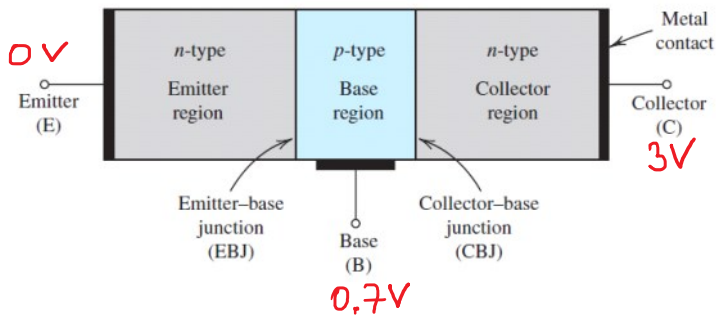
from Q1

Diodes is ideal. $V_o = ?$

$V_I < 0$ thus the diode will be off.

$$V_o = V_I$$

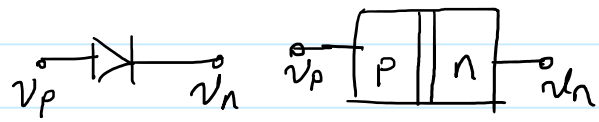
Q3



In which region the BJT is operating?
cutoff / active / saturation

Table 6.1 BJT Modes of Operation

Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward



$v_p \geq v_n \Leftrightarrow$ Forward

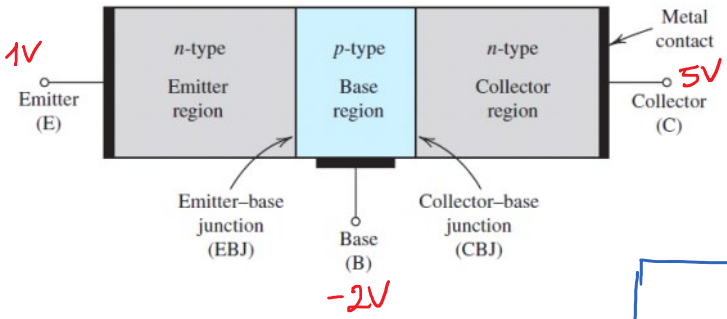
$v_p < v_n \Leftrightarrow$ Reverse

(EBJ) $V_B - V_E = v_p - v_n = 0.7$ Forward

(CBJ) $V_B - V_C = v_p - v_n = -2.3$ Reverse

Active Mode

Q3

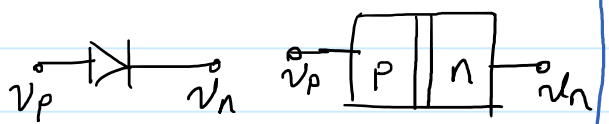


In which region the BJT is operating?
cutoff / active / saturation

Table 6.1 BJT Modes of Operation

Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward

Ideal Diode Model



$v_p > v_n \Leftrightarrow$ Forward

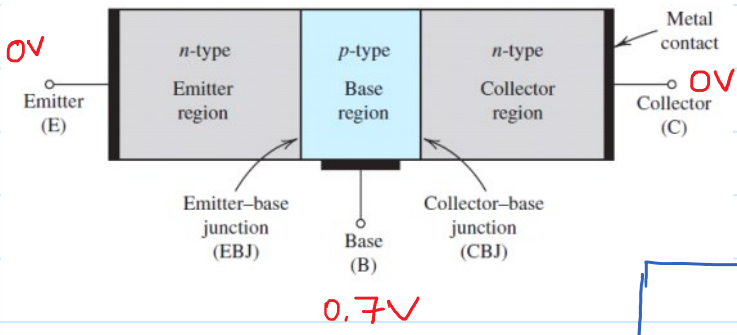
$v_p < v_n \Leftrightarrow$ Reverse

EBJ $V_B - V_E = -2 - 1 = -3$ Reverse

CBJ $V_B - V_C = -2 - 5 = -7$ Reverse

Cutoff

Q3

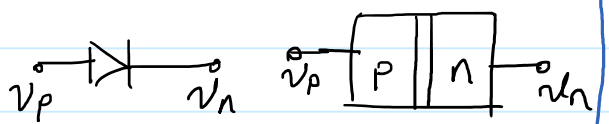


In which region the BJT is operating?
cutoff / active / saturation

Table 6.1 BJT Modes of Operation

Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
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Ideal Diode Model



$v_p > v_n \Leftrightarrow$ Forward

$v_p < v_n \Leftrightarrow$ Reverse

EBJ $V_B - V_E = v_p - v_n = 0.7 \Rightarrow$ Forward

CBJ $V_B - V_C = v_p - v_n = 0.7 \Rightarrow$ Forward

Saturation